

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396439
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Miller; Simon G.

Connector

Abstract

A connector has a first member (1) forming a female member, and a second member (2) forming a male member arranged to co-operate with the female member along a longitudinal axis (3). The first member houses a spring biased claw (44) which has a finger (49) arranged to engage in a detent (25) in the second member so as secure the second member to the first member. A latch (70), which is reciprocal in a direction perpendicular to the longitudinal axis (3), is arranged to abut the claw (44) to lock and prevent the claw from opening when engaged in the detent (25). A magnet (76) is attached to the latch (70) and the latch (70) is spring biased by a spring (78) to lock the claw. A second magnet (590,1080) is positioned adjacent the first magnet (76) to withdraw the latch from engagement with the claw, thereby enabling removal of the second member (2).

Inventors:	Miller; Simon G. (Shoeburyness, GB)
Applicant:	Rawlings; Paul Simon (Chelmsford, GB)
Family ID:	1000008780573
Assignee:	Magron Ltd (Brasildon, GB)
Appl. No.:	18/278971
Filed (or PCT Filed):	March 02, 2022
PCT No.:	PCT/GB2022/050542
PCT Pub. No.:	WO2022/185047
PCT Pub. Date:	September 09, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240147966 A1	May. 09, 2024

Foreign Application Priority Data

Publication Classification

Int. Cl.: A01K27/00 (20060101); A44B11/25 (20060101)

U.S. Cl.:

CPC A01K27/005 (20130101); A01K27/001 (20130101); A01K27/002 (20130101);
A44B11/2592 (20130101); A44D2201/32 (20130101); A44D2203/00 (20130101)

Field of Classification Search

CPC: A01K (27/005); A01K (27/001); A01K (27/002); A44B (11/2592); A44D (2201/32);
A44D (2203/00); F16B (2/10); F16B (2200/83)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2007/0134061	12/2006	Nance	403/DIG.1	E05B 47/0045
2018/0343834	12/2017	McConnell	N/A	A01K 1/04
2021/0368949	12/2020	Prasitkittanai	N/A	A44C 5/20
2021/0392856	12/2020	Miller	N/A	A44B 11/2534
2023/0172325	12/2022	Selbe	24/658	A44C 5/2052

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2511104	12/2013	GB	N/A
WO2004088151	12/2003	WO	N/A

Primary Examiner: Upchurch; David M

Attorney, Agent or Firm: Weiss & Moy, PC

Background/Summary

(1) This invention relates to a connector and particularly, although not exclusively, to a quick release connector for an animal harness.

(2) Although the connector of this invention has particular efficacy as a connector for an animal harness, the invention is not limited thereto and it is envisaged that the connector may be used in general where two parts are to be connected together, such as a chest strap on a backpack or a diving compressed air canister harness.

(3) It is known for an animal collar, such as a collar for a cat or dog, to include a loop that is usually made of metal material and for the loop to be attachable by a spring clip to a lead. Such a spring clip on a lead is difficult to operate for people with rheumatism or arthritis in their hands, since they are unable to apply sufficient pressure to actuate the clip.

(4) One known magnetic padlock is the Genii magnetic lock and another is the Maverick magnetic lock, which are magnetically operable combination locks.

(5) GB-A2511104 discloses a female body defining a receiving chamber for a male engaging member. A magnet is located inside the female member and the male member has a metal insert so that the male member is magnetically drawn into the female member. A pair of spigots are pivotally attached to the female member and arranged to be manually operated to move the spigots into and out of engagement with a circumferential groove in the male member to lock the male member to the female member. Opposing ends of the male and female members are provided with attachment hooks such that the hook on the male member is attachable to a dog collar and the hook on the female member is attachable to a lead. Such a connector facilitates ready attraction between the male and female members, but suffers from the same disadvantage as a conventional spring clip in that the spigots are manually operable and those with rheumatism, or arthritis or similar hand impairment, find unlocking the spigots so that the male and female members may be moved apart, difficult.

(6) The present invention seeks to provide a connector in which the foregoing problems are substantially mitigated.

(7) According to a first aspect of this invention, there is provided a connector comprising a first member forming a female member and a second member forming a male member arranged to cooperate with said female member along a longitudinal axis, said first member housing spring biased claw means having at least one claw arranged to engage a detent in said second member, and spring biased latch means arranged to prevent opening of said claw means, said latch means further including at least one first magnetic member to facilitate said latch means to disengage said detent against said spring bias, whereby, in operation, when said second member is moved in a first direction into said first member, it initially raises said latch means against the spring bias thereof, further motion in said first direction opens the at least one claw, continued motion in said first direction causes the substantially simultaneous operation of the claw to move into said detent under the spring bias thereof and said latch means to move into contact with said claw means under the spring bias thereof to prevent said at least one claw from opening until the latch is lifted by a second magnetic member acting in cooperation with said at least one first magnetic member to lift the latch means out of said contact and the second member may be removed from the first member in a second direction opposite to said first direction.

(8) Preferably, the first and second members are arranged to be releasably and removably interconnected with one another, said first member having a hollow cavity extending along said longitudinal axis so that a first portion of said cavity is arranged to receive said second member along said longitudinal axis, a second portion of said cavity locating, along said longitudinal axis, the claw means so that the at least one claw is spring biased toward said longitudinal axis, and a spring biased pusher is arranged to act against a distal end of said second member, wherein when said second member is inserted into the cavity in the first direction, it moves the pusher against the spring bias of the pusher and opens the claw means against the spring bias of the claw means until the claw is moved under spring bias pressure thereof to be engageable in the detent of the second member and, when engaged, the latch means moves into contact with the claw means to prevent the claw means from opening until the latch is lifted by the second magnetic member out of said contact, and said pusher moves said second member in the second direction opposite to the first direction.

(9) Conveniently, the spring bias for the claw means and for the pusher is provided by a common compression spring, i.e., a spring common to both the claw means and the pusher.

(10) Preferably, the claw means comprises at least two cantilevered claws which are each arranged to pivot about an axle having an axis substantially perpendicular to said longitudinal axis, each said claw being U-shaped, with the base of the U-shape extending in the plane of said longitudinal axis, a proximal end of said base having an arm extending therefrom and said axle being located

substantially at a junction of the base and the arm, a distal end of said base having the claw extending towards said longitudinal axis for engaging in said detent.

(11) Conveniently, two cantilevered claws are provided diametrically opposed to one another.

(12) Advantageously, a proximal end of the arm is provided with a recessed notch on a remote side of the axle to the base of the U-shape to bias the claw inwardly toward the longitudinal axis and an opposite end of the common compression spring being located in a cavity formed in the pusher so that the common compression spring has the dual operation of biasing the distal end of the claw toward the longitudinal axis and biasing the pusher in said second direction.

(13) Preferably, the latch means includes an orthogonally extending member which extends substantially perpendicularly to the longitudinal axis and at a first end of said longitudinal extending member is an inverted U-shaped member having a pair of opposed arms which extend in the same plane as the longitudinally extending member, the arms being arranged to contact a respective claw of the claw means to prevent the claw means from opening when the second member is fully inserted into the first member.

(14) Alternatively, the latch means includes a pair of opposed pivotable arms arranged on pivot means, said arms being biased by a compression spring so that the arms are biased over an outside surface of each U-shaped claw to lock the U-shaped claws at their innermost extent until the latch is released by the second member.

(15) Advantageously, at an end of the orthogonally extending member remote from the inverted U-shaped member is a body housing the at least one first magnetic member and which is contacted by one end of a compression spring, a remote end of the compression spring engaging with an undersurface of a cap, whereby the compression spring biases the latch means in a direction substantially orthogonal to said longitudinal axis and toward said longitudinal axis.

(16) In a preferred embodiment, said first magnetic member comprises at least one real earth magnet and, advantageously, six real earth magnets are used.

(17) Advantageously, the second member has an inwardly tapering distal end which is chamfered to raise the latch means against the spring bias thereof.

(18) Conveniently, the second member has a circularly cross-section body and the detent is formed by a circumferential groove adjacent to said distal end and a leading edge of said groove has a chamfer to cooperate with a corresponding chamfer on an inside surface of the claw to facilitate removal of the second member from the claw means.

(19) Advantageously, a proximal end of the second member has an aperture for attachment of utilization means such as an animal lead.

(20) Conveniently, a housing provides support for each axle of the cantilevered claws and at an end of the housing remote from the claw means is a first connecting member in which is provided an aperture for attachment to another utilization means such as an animal collar.

(21) According to a second aspect of this invention there is provided a method of operating a connector, as claimed in any of the preceding claims, including the steps of: a) moving the second member into the first portion of the cavity of the first member to lift the latch means against the spring bias thereof and to push against the spring bias of the pusher and to open the at least one claw of the claw means until the at least one claw moves into the detent in the second member and the latch means is biased by the spring thereof to cause the arms of the inverted U-shaped member to move to prevent the claw means from opening, b) providing the second magnetic member to withdraw the latch means against the bias of the spring thereof, whereupon the at least one claw is permitted to open and the pusher moves the second member in the second direction.

(22) According to a third aspect of this invention there is provided a connector in accordance with said first aspect in combination with a harness or webbing or animal lead or support device.

Description

(1) The invention will now be described, by way of example, with reference to the accompanying drawings, in which:

(2) FIG. 1 shows an exploded perspective view of a connector in accordance with this invention,

(3) FIG. 2 shows a vertical cross-section of the connector before insertion of the second member into the first member,

(4) FIG. 3 is a vertical cross-section view showing the second member raising the latch and abutting the pusher,

(5) FIG. 4 shows a vertical cross-section of the second member locked into position in the first member,

(6) FIG. 5 shows an exploded upper side perspective view of a further embodiment of a connector in accordance with this invention,

(7) FIG. 6 shows an exploded underside perspective view of the further embodiment of the connector shown in FIG. 5,

(8) FIG. 7 shows a vertical cross-section of the connector of FIGS. 5 and 6 before insertion of the second member into the first member,

(9) FIG. 8 shows a vertical cross-section view of the connector shown in FIGS. 5 and 6 with the second member raising the latch and abutting the pusher,

(10) FIG. 9 shows a vertical cross-section of the connector shown in FIGS. 5 and 6 with the second member locked into position in the first member,

(11) FIG. 10 shows an exploded perspective view of another embodiment of a connector in accordance with this invention when viewed from above,

(12) FIG. 11 shows an exploded perspective view of the another embodiment of the connector shown in FIG. 10 when viewed from below,

(13) FIG. 12, shows a vertical cross-section view of the connector shown in FIGS. 10 and 11 before insertion of the second member into the first member,

(14) FIG. 13 shows a vertical cross-section view of the connector shown in FIGS. 10 and 11 with the second member raising the latch and abutting the pusher, and

(15) FIG. 14 shows a vertical cross-section of the connector shown in FIGS. 10 and 11 with the second member locked into position in the first member.

(16) In the Figures like reference numerals denote like parts.

(17) The connector shown in the FIGS. 1-4 has a first member 1 and a second member 2, each disposed about a longitudinal axis 3.

(18) The second member 2, in the currently preferred embodiment, has a circularly cross-sectioned body 21 having a proximal end 22 and a distal end 23. The distal end has an inwardly tapered head 24, for reasons that will be explained hereinafter, and a detent in the form of a circumferential groove 25, the leading edge 26 of the groove being chamfered, again, for reasons that will be explained hereinafter. The proximal end of the second member has opposed flat surfaces 27 through which is formed an aperture 28 for connection of utilization means such as an animal lead.

(19) The first member 1 has a housing in two parts, an upper part 10, as shown in the drawings, and a lower part 11, as shown in the drawings, and longitudinally extending through the housing is a circular cross-section cavity 12 for cooperatively mating with the body of the second member 2. The cavity 12 has a first diameter 13, which is an easy sliding fit with the diameter of the body 21 and head 23, and an intermediate diameter cavity 14 in which is located a circularly cross-sectioned pusher 30, the confluence of the cavities 13 and 14 acting as a stop 15 for the pusher. The upper part 10 and the lower part 11 may be connected together by any suitable means such a screws, nuts and bolts, welding or adhesive.

(20) Remote from the cavity 12 is a generally rectangularly shaped cavity 16, into which is fixedly located a casing 40 having an aperture 41 at one end thereof, and at an opposite end of the aperture 41 is a connecting member 42 having an aperture 43 which may be connected to, for example, a

utilization means such as an animal collar.

(21) Located in the aperture **41** are a pair of opposed U-shaped claws **44** on each side of the longitudinal axis **3** and each having a generally rectangular cross-section. Each claw has a proximal end **45** with an arm **46** having a hole **47** therethrough for location of an axle which passes into holes **48** in opposing faces of the aperture **41** in the housing **40**. An inner face of each arm on a remote side of the axle from the base of the U-shape of the claw, ie nearer to the longitudinal axis, has a notch **59** against which acts one end **61** of a common spring **60**, which is a compression spring.

(22) The distal end of each claw has a finger **49** directed inwardly towards the longitudinal axis. An inner face **50** of each finger has a corresponding chamfer to the chamfer on the leading edge **26**.

(23) The pusher **30** is cylindrical having an open end and a closed end **31**. A second end **62** of the common spring **60** is inserted into the open end to abut the inside of the closed end **31**. In operation, the common spring **60**, acts to bias the fingers **49** inwardly towards the longitudinal axis and biases the pusher **30** in a second direction which is opposite to a first direction that is the direction of insertion of the second member **2** into the first member **1**.

(24) The upper part **10** of the housing has an aperture **17** therethrough extending orthogonally to the longitudinal axis **3**.

(25) The aperture **17** is counter bored to provide an inner circularly cross-sectioned hole **18** and an outer circularly cross-sectioned hole **19**, the hole **19** being of larger diameter than the hole **18**.

(26) Mounted in the holes **18** and **19** is a latch **70** formed by an inverted U-shaped member **71** having a base **72** and depending arms **73**. Opposite to the arms **73** is a circularly cross-sectioned member **74** extending orthogonally to the longitudinal axis and which is located in the hole **18**. The member **74** is partially located inside a circularly cross-sectioned body **75**, and an end of the body **75** remote from the member **74** contains, in the preferred embodiment, six rare earth magnets forming a first magnet **76**. An upper end of the body **75** has a lip **77** against which acts a compression spring **78** and an end of the compression spring **78** remote from the body **75** acts against an inside surface of a cap **79** which is securely located in the hole **19** and houses the body **75**. It will be understood by those skilled in the art that the cap may be integrally moulded with the upper part **10**.

(27) Thus, the casing **40** and the cap **79** are fixedly located in the first member **1** with the plunger **30** being reciprocal in the direction of the longitudinal axis and the body **75**, and rare earth magnets **76** are reciprocal in a direction transverse to the longitudinal axis.

(28) In operation, as shown in FIG. 2, the second member **2** is inserted into the first member **1** in the first direction. As the member **2** moves into the cavity **13**, so it raises the latch **70** and it is for this reason that the tapered head **24** is provided. As shown in FIG. 3, further movement of the member **2** into the member **1** cause the tapered head **24** to abut the closed end **31** of the pusher **30** to compress the common spring **60** and, at substantially the same time, for the tapered head **24** to act against a remote end of the fingers **49** so that the fingers are opened outwardly pivoting about their respective axles and against the bias of the common spring **60** located in the notches **59**. Referring to FIG. 4, the movement of the second member **2** is complete when the fingers **49** move under the bias of the common spring **60** into the groove **25** and the latch moves toward the longitudinal axis **3**. The inside edge of the arms **73** act against the outer edge of the claws **40** to prevent the claws from opening so that the fingers **49** that are located in the groove **25** securely lock and prevent removal of the member **2** from the member **1**.

(29) To unlock the second member **2** from the first member **1**, i.e., to move the second member **2** out of the first member **1** in the second direction, which is opposite to the first direction, a second magnetic member **590,1080** (not shown in FIGS. 1-4, but depicted in FIGS. 5-14) is positioned on or closely adjacent to the cap **79**, having an opposite polarity to that displayed by the upper face of the first magnet **76**. Thus, the latch **70**, which is connected to the body **75** is moved in a direction away from the longitudinal axis to lift the latch **70** out of contact with the claws **44**. The claws **44**

are, thus, released so that the fingers **49** may move outwardly away from the longitudinal axis when the second member **2** is pulled in the second direction by virtue of the mating chamfer **26** and the corresponding chamfer on the inner face **50** of each finger.

(30) Although the cavity **12**, in a preferred embodiment, has a circular cross-section, it could be, for example, a cuboid-shaped such as a rectangular cavity. So as to mate with the cuboid-shaped cavity, the body of the second member **2** may have a suitable mating cuboid-shaped cross-section.

(31) It will, thus, be appreciated the connector thus described forms a quick-release mechanism that may be used for an animal lead to connect to a collar on the animal and, provided that a person is able to hold a magnet, then they do not need to exert any physical force to effect release and only a minor force to operate against the common spring **60**.

(32) In one use of the invention, the connecting member **42** may be attached to an animal collar. In such a use, it is envisaged that once the connecting member **42** is attached to the animal collar the first member **1** will remain attached to the collar and that once the second member **2** is attached to an animal lead, then it will also remain attached thereto. Thus, the connector of this invention may be sold fixedly secured to a lead, or as a device that may be connected to a lead as an aftermarket device.

(33) A further connector as hereinbefore described with or without connecting member **42** may be secured to the proximal end of the second member **2** at aperture **28** or instead of flat surfaces **27** and aperture **28**. Similarly, a connector as hereinbefore described may be secured to the casing **40** with or without connecting member **42**.

(34) With an increase in dog thefts, especially from outside shops, it is thought that the present invention could help mitigate this crime.

(35) However, it is envisaged that the connector of this invention has greater utilization than as an animal lead connector, especially where a user is manually impaired through physical, medical impairment, or even when a user's hands and fingers are capable of little or no movement through extreme cold.

(36) Although shown as having holes **28** and **43** for connection to utilization means, any suitable means of connection may be provided to other items.

(37) FIGS. **5** and **6** show exploded perspective views of a further embodiment of a connector in accordance with this invention having a different form of latch. The lower part **11** has an internal cavity **516** for locating the claws **44** and a latch **570**. In this respect, instead of the latch being an inverted U-shaped member as in FIGS. **1-4**, in the embodiment of FIGS. **5-9**, a latch **570** which is generally U-shaped is located over and on top of the claws **44**.

(38) A rearward end of the latch **570** remote from opposed arms **573** has depending wings **571** which each have an aperture about which the latch pivots around an axle **572**. Fixedly secured between the arms **573** is a body **575** for location of the rare earth magnets forming the first magnet **76** and the compression spring **78** acting between the body **575** and an underside of the upper part **10**. The body has a depending limb **576** between the arms **573**. The upper part **10** of the first member **1** has, in a preferred embodiment, an integrally moulded cap **579**.

(39) Each of the U-shaped claws **44** has a respective axle **580** passing through the hole **47** and the opposite ends of each axle **580** are secured in upper part **10** and lower part **11** respectively.

(40) The pusher **30** has diametrically opposed lugs **531** which are arranged to abut depending limb **576** on an underside portion of the body **575** that is located below the base of the U-shape of the latch **570**, and in a groove **700** in the inside of the cavity **14**, as particularly shown in FIG. **7**.

(41) A second magnetic member **590** has a body **591** for securing (in the present embodiment) seven rare earth magnets **592** and the body has an apertured lug **594** for attachment to a chain or the like.

(42) Prior to insertion of the second member into the first member, the compression spring **78** forces the body **575** and, hence latch **570**, in a downwardly direction (as viewed in the drawings) so that each of the arms **573** overlaps a respective claw **44** to hold the latch in a closed position.

(43) In operation, when the second member is inserted in the first direction into the first member, as shown in FIG. 7, the distal end **23** is located in the cavity **12** and the lower lug **531** abuts the groove **700** under the pressure of common spring **60**; the upper lug **531** (as shown in the drawings) being restrained by the latch **570** and the lower lug **531** being restrained by the groove **700** in the cavity **14** in the lower part **11**. In this initial position, the arms **573** restrain the U-shaped claws **44** from opening. As shown in FIG. 8, further movement of the member **2** into the member **1** causes the tapered head **24** to abut an underside of the body **575** which is raised against the pressure of the compression spring **78**. This causes the latch to pivot about axle **572** and lift arms **573** out of contact with the outer side of the U-shaped claws **44**. Further insertion of the second member causes the distal end **23** of the second member to abut the pusher **30** to compress common spring **60** to cause the fingers **49** of the U-shaped claws **44** to open by pivoting about axles **580**.

(44) As shown in FIG. 9, further movement of the second member into the first member is complete when the depending limb **576** is pushed downwardly (as viewed in the drawings) by compression spring **78** to locate into the groove **25** and the arms **573** locate over an outer side of each U-shaped claw **44** to lock the fingers **49**, which are biased inwardly by the force of the common spring **60** into the groove **25**. Thus, simultaneously with the fingers moving into the groove **25**, so the arms **573** move downwardly to lock the claws from opening and the depending limb **576** also moves into the groove **25** to securely lock member **2** into member **1**.

(45) As in the embodiment of FIGS. 1-4, the second member is unlocked from the first member by the application of a magnetic force from the second magnetic member **590** to raise the body **575** and, hence latch **570**, to permit the claws and depending limb **576** to open out of the groove **25**.

(46) Referring now to FIGS. 10-14, there is shown another embodiment of the connector of this invention which is less bulky and which has a second member of a flat, key-like form, that is generally cuboid. The second member **1002** is arranged to be inserted into the first member **1001** formed of an upper (as shown in the drawings) first part **1010** and a second lower part **1011**.

(47) The second member **1002** has a finger grip portion **1022** having a width which is larger than an entrance **1013** of a cavity **1014** in the lower part **1011**. A body **1003** of the second member **1002** is narrower than the finger grip portion **1022** so as to fit into the entrance **1013**. The body, at an end remote from the finger grip portion **1022**, has chamfered outer edges **1004** and along a minor side of the body **1003** are a pair of opposed indents **1005**. Extending through major planar faces of the body **1003** is a rectangular aperture **1006**.

(48) The pusher **30** has a boss **1031** against which a front face **1007** of the second member abuts, in operation.

(49) The latch **1070** has a pair of wings, each with a depending wing **1071**, each having a hole therethrough for the location of a respective pivot pin **1072** extending through a hole in a side wall of the lower part **1011**. On a remote end of the latch **1070** from the wings **1071** are a pair of opposed depending arms **1073** located over the claws **44** and arranged, in operation, to lock the latch in an inwardly direction.

(50) The body **575** locating the magnet **76** is biased by a compression spring **78** (not shown in FIG. 10).

(51) In this another embodiment, a second magnetic member **1080** has a body **1081** for securing (in the present embodiment) seven rare earth magnets **1082** and the body has a cap **1083** having an apertured lug **1084** for attachment to a chain or the like.

(52) The operation of the another embodiment will now be described with reference to FIGS. 12-14.

(53) In FIG. 12, the second member is entering the entrance **1013** and the latch is pivoted by the force of the common spring **60** into a closed position, whereby the fingers **49** are biased inwardly and the fingers **49** are held closed by the arms **1073** of the latch being pushed downwardly over the outside of the U-shaped claws by the compression spring **78**.

(54) Relative movement of the second member **1002** into the first member **1001** causes the front

face **1007** to abut the boss **1031** and to start to compress the common spring **1060**. Simultaneously, the second member **1002** abuts the depending limb **576** to cause the compression spring **78** to compress and the arms **1073** to be lifted out of contact from the outside of each U-shaped claw **44**. Further compression of the common spring **60** causes the U-shaped claws to open by pivoting about their respective axles **580**.

(55) As shown in FIG. **14**, further insertion of the second member **1002** into the first member **1001** causes depending limb **576** to drop under the force of the compression spring **78** into the rectangular aperture **1006** and for the latch to rotate under the force of the compression spring **78** so that the respective arms **1073** move into the opposed indents **1005**. The second member is, thus, securely locked into the first member.

(56) Unlocking of the second member from the first member is occasioned by the second magnetic member interacting with the magnet **76** so that the body **575** and the arms **1073** are raised out of contact with the U-shaped claws and the common spring **60** causes the U-shaped claws to open out of contact from opposed indents **1005** to permit the second member to be withdrawn.

(57) The material for the components of the connector may be made from any suitable materials such as plastics, aluminium or stainless steel.

Claims

1. A connector comprising a first member (**1**) forming a female member and a second member (**2**) forming a male member arranged to cooperate with said female member along a longitudinal axis (**3**), said first member housing a spring (**60**) for biasing a claw means (**44-49**) having at least one claw (**44**) arranged to engage a detent (**25**) in said second member, and a further spring (**78**) for producing a spring bias applied to a latch means (**70-74**), said latch means including at least one arm (**73**) arranged to engage over an outer surface of said at least one claw (**44**) arranged to prevent opening of said claw means, said latch means further including at least one first magnetic member (**76**) to facilitate said latch means to disengage said detent against said spring bias, whereby, in operation, when said second member (**2**) is moved in a first direction into said first member (**1**), said second member initially raises said latch means (**70-74**) against the applied spring bias of the further spring (**78**), further motion in said first direction opens the at least one claw (**44**), continued motion in said first direction causes the simultaneous operation of the claw (**44**) to move into said detent (**25**) under the bias of said spring (**60**) and said latch means (**70-74**) to move into contact with said claw means (**44-49**) under the further spring (**78**) bias thereof to prevent said at least one claw (**44**) from opening until the latch is lifted by a second magnetic member (**590,1080**) acting in cooperation with said at least one first magnetic member (**76**) to lift the latch means out of said contact and the second member is removed from the first member in a second direction opposite to said first direction.

2. A connector as claimed in claim 1, wherein the first (**1**) and second (**2**) members are arranged to be releasably and removably interconnected with one another, said first member having a hollow cavity (**14**) extending along said longitudinal axis so that a first portion of said cavity is arranged to receive said second member along said longitudinal axis, a second portion of said cavity locating, along said longitudinal axis, the claw means so that the at least one claw is spring biased toward said longitudinal axis, and said spring (**60**) also producing a spring bias on a pusher (**30**) which is arranged to act against a distal end (**23**) of said second member, wherein when said second member is inserted into the cavity in the first direction, so said second member moves the pusher against the spring bias of the spring (**60**) on the pusher and opens the claw means against the spring bias of said spring (**60**) on the claw means until the claw is moved under spring bias pressure thereof to be engageable in the detent (**25**) of the second member and, when engaged, the latch means moves into contact with the claw means to prevent the claw means at least one claw from opening until the latch is lifted by the second magnetic member out of said contact, and said pusher moves said

second member in the second direction opposite to the first direction.

3. A connector as claimed in claim 2, wherein the spring bias for the claw means and for the pusher is provided by a common compression spring (60).

4. A connector as claimed in claim 1, wherein the claw means comprises at least two cantilevered claws (44) which are each arranged to pivot about an axle having an axis substantially perpendicular to said longitudinal axis (3), each said claw being U-shaped, with the base of the U-shape extending in the plane of said longitudinal axis, a proximal end (45) of said base having an arm extending therefrom and said axle being located substantially at a junction of the base and the arm, a distal end of said base having each said claw extending towards said longitudinal axis for engaging in said detent.

5. A connector as claimed in claim 4, wherein two cantilevered claws are provided diametrically opposed to one another.

6. A connector as claimed in claim 4, wherein a proximal end (45) of the arm is provided with a recessed notch (59) on a remote side of the axle to the base of the U-shape to bias the claw inwardly toward the longitudinal axis and an opposite end of the common compression spring (60) being located in a cavity formed in the pusher (30) so that the common compression spring has the dual operation of biasing a distal end of each said claw toward the longitudinal axis and biasing the pusher in said second direction.

7. A connector as claimed in claim 1, wherein the latch means (70) includes an orthogonally extending member (74) which extends substantially perpendicularly to the longitudinal axis and at a first end of said longitudinal extending member (74) is an inverted U-shaped member (71) having a pair of opposed arms (73) which extend in the same plane as the longitudinally extending member, the arms being arranged to contact a respective claw (44) of the claw means to prevent the claw means from opening when the second member (2) is fully inserted into the first member (1).

8. A connector as claimed in claim 1, wherein the latch means includes a pair of opposed pivotable arms (573, 1073) arranged on pivot means (572, 1072), said arms being biased by the further spring (78) so that the arms are biased over an outside surface of each U-shaped claw (44) to lock the U-shaped claws to an innermost extent of the U-shaped claws until the latch is released by the second member.

9. A connector as claimed in claim 8, wherein there is provided a body (75) housing the at least one first magnetic member (76) and which is contacted by one end of the further spring (78), a remote end of the further spring engaging with an undersurface of a cap (79), whereby the further spring biases the latch means in a direction substantially orthogonal to said longitudinal axis and toward said longitudinal axis.

10. A connector as claimed in claim 9, wherein said first magnetic member comprises at least one real earth magnet.

11. A connector as claimed in claim 1, wherein the second member (2) has an inwardly tapering distal end (24) which is chamfered to raise the latch means against the spring bias thereof.

12. A connector as claimed in claim 1, wherein the second member has a circularly cross-section body and the detent (25) is formed by a circumferential groove adjacent to said distal end and a leading edge of said groove has a chamfer (26) to cooperate with a corresponding chamfer on an inside surface of the claw to facilitate removal of the second member from the claw means.

13. A connector as claimed in claim 1, wherein a proximal end of the second member has an aperture (28) for attachment of utilization means.

14. A connector as claimed in claim 4, wherein a housing (40) provides support for each axle of the cantilevered claws and at an end of the housing remote from the claw means is a first connecting member (42) in which is provided an aperture (43) for attachment to another utilization means.

15. A connector in accordance with claim 1 attached to one of a harness, webbing, animal lead, and a support device.

16. A connector as claimed in claim 13, wherein the utilization means is an animal lead.

17. A connector as claimed in claim 14, wherein the utilization means is an animal collar.
